# Capstone Project Introduction

ECON 490

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# Overview

In these slides, we'll talk about:

- Requirements for the capstone project
- Basic components of your projects
- Tips for your proposals and common data analysis pitfalls

# Key Components of Your Capstone Papers

Core components of your capstone projects:

1. Clear research question and motivation
2. Data analysis

We'll complete several intermediate assignments throughout the semester

- First step is the Research and Data Proposal
- Capstone Outline will include draft data analysis and presentations

Proposal/Outline assignments have detailed templates for you to use/submit

# Capstone Project Grade Breakdown

The capstone project is worth ***30%*** of your final grade

| ***Assignment*** | ***Due Date*** | ***% of Final Grade*** |
|---|---|---|
| Research Proposal | Next Monday | 2.5% |
| Outline | Monday, 4/13 | 5% |
| Outline Presentation | Week of Monday, 4/13 | 2.5% |
| Final Presentation | Weeks 14/15 | 10% |
| Final Paper | End of Finals Week | 10% |

# Final Deliverables

All capstone output should be:

- Written in your own words
- Neatly-formatted/presentable (no R screenshots unless specifically requested)

**Final Papers:** Roughly 8-10 pages

- Including data analysis output and works cited
- We will discuss writing and paper requirements later in semester

**Presentations**

- *Final Presentations:* ~15 minutes
- *Outline Presentations:* 5-10 minutes

# Capstone Research and Data Proposal

First capstone assignment is due Week 10 (3/23)

Steps to complete this assignment:

1) Decide if you want to work alone or with a partner
2) Identify a clear research question
3) Identify the data you'll use to answer your research question
4) Complete the proposal worksheet on Canvas

***NOTE:*** *Before moving forward with your project, you will need approval on the topic and data source(s) – I may ask you to submit a revised proposal.*

# Working Backwards

Trying to find a research question can be daunting

To make things easier, try ***working backwards***

- Start thinking about the general topics you're interested in – this could be labor markets, housing, education, poverty, etc.
- Then ask, "What kind of data analysis can I do?" related to that topic

Your data analysis requires (1) data and (2) an analysis strategy

- Start by identifying and exploring the data set that best fits your interest
- Then, think about (2) possible analysis strategies
- Finally, return to research question and orient around your analysis

# General Suggestions on Proposals

Key first step—find and download data (ideally multiple data sets)

- Load data in R, read documentation (!), run simple regressions
- Ask yourself, “Does this data look like something I can work with?”

Your first draft of a research question will likely be too superficial

- Remember, the goal is to write an 8+ page paper
- As part of this, you’ll spend 2-3 pages talking about your data analysis output

*If your research question is something like, “What’s the correlation between GDP and unemployment?” you’re not going to have much to say!*

# Selection of Research Questions from Prior Semesters

*How do recessions impact household food expenditures?*

*How do the returns to graduate education vary across demographic groups?*

*How did the Covid-19 pandemic impact household transportation choices?*

*What factors drive variation in home prices across counties in California?*

*How much of NBA player compensation can be explained by future expected performance (compared to past performance or prior accolades)?*

# Thinking Descriptively

How to go beyond, “What’s the correlation between X and Y?”

- For ECON 340, running 1-2 regressions is generally sufficient for a paper
- In 490, goal is to concretely answer a question/say something interesting

In general, think about ***exploring variation*** in either:

1. The level of one economic variable as a function of other variables
2. A relationship between economic variables

In both cases, you can explore variation across time, places, economic factors, etc.

# **Data Analysis Pitfalls** – *Macro Time Series Data*

*General rule of thumb*: make sure you have data for multiple groups

- In a lot of data sets/research contexts, this happens naturally
- People in different states, players on different NBA teams, etc.

Avoid data for just a single place/company/person over time

- Lack of groups = lack of natural comparisons to make via regression
- Severely limits flexibility in your analysis = hard to write a 10-page paper

Want to do a macro topic? Look up data for multiple countries, or states in US

# **Data Analysis Pitfalls** – *Ecological Fallacy*

A common 490 topic that I've decided to prohibit: *"What is the correlation between state economic conditions and crime?"*

Two problems:

1) Correlations are generally a bit weak
2) Correlations are very easy to misinterpret

Ecological fallacy = making inferences about individuals based on group-level data

- Asking, "How does losing my job affect my propensity to commit crime" is very different than asking, "How do recessions impact crime?"

# **Data Analysis Pitfalls** – *Getting Too Fancy*

In general, OLS should be fully sufficient for most projects

- If you have a background/prior course experience with ML/time series metrics/etc., you are welcome to apply these tools!
- However, don't let AI push you into using techniques you haven't studied

Using what feels like a fancy method doesn't make your answer any better if the question isn't well phrased or the data isn't well-suited to your analysis

ChatGPT loves logit/probit regression and variance inflation factors

- Avoid these tools unless you 1) know why you're using them and 2) know how to correctly interpret their output